

Task 1: Python File Handling And Automation

Understand Python file handling, automation logic, and exception handling.

Requirements

- Read and write files (txt/csv)
- Automate file operations (rename, move, delete)
- Use try-except for error handling
- Explain code logic in comments

Deliverables

1. Python script with proper comments
2. Sample input/output screenshots

Code Snippet:

Filename: main.py

Import required modules

import os

import shutil

Function to read a text file

def read_file(filename):

try:

with open(filename, "r") as file:

content = file.read()

print("\nFile Content:")

print(content)

except FileNotFoundError:

print("Error: File not found")

except Exception as e:

print("Something went wrong:", e)

```
# Function to write data into file
```

```
def write_file(filename, text):
```

```
    try:
```

```
        with open(filename, "a") as file:
```

```
            file.write(text)
```

```
        print("\nData written successfully")
```

```
    except Exception as e:
```

```
        print("Error while writing:", e)
```

```
# Function to rename file
```

```
def rename_file(old_name, new_name):
```

```
    try:
```

```
        os.rename(old_name, new_name)
```

```
        print(f"\nFile renamed from {old_name} to {new_name}")
```

```
    except FileNotFoundError:
```

```
        print("Rename Error: File not found")
```

```
    except Exception as e:
```

```
        print("Rename Error:", e)
```

```
# Function to move file
```

```
def move_file(source, destination):
```

```
    try:
```

```
        shutil.move(source, destination)
```

```
        print(f"\nFile moved to {destination}")
```

```
    except FileNotFoundError:
```

```
        print("Move Error: File not found")
```

```
except Exception as e:  
    print("Move Error:", e)
```

```
# Function to delete file
```

```
def delete_file(filename):  
    try:  
        os.remove(filename)  
        print(f"\n{filename} deleted successfully")
```

```
except FileNotFoundError:  
    print("Delete Error: File not found")
```

```
except Exception as e:  
    print("Delete Error:", e)
```

```
# ----- MAIN PROGRAM ----- #
```

```
print("==== Python File Handling & Automation ====")
```

```
# Step 1: Write into file
```

```
write_file("sample.txt", "\nWelcome to Python Internship Task")
```

```
# Step 2: Read file
```

```
read_file("sample.txt")
```

```
# Step 3: Rename file
```

```
rename_file("sample.txt", "updated_sample.txt")
```

```
# Step 4: Create folder if not exists
```

```
folder_name = "files"
```

```
if not os.path.exists(folder_name):
    os.mkdir(folder_name)
    print("\nFolder created:", folder_name)

# Step 5: Move file into folder
move_file("updated_sample.txt", "files/updated_sample.txt")

# Step 6: Delete file example
delete_file("test.csv")

print("\nAutomation Task Completed")
```

output:

```
E:\alfido py dev\project>python main.py
===== Python File Handling & Automation =====

Data written successfully

File Content:

Welcome to Python Internship Task

File renamed from sample.txt to updated_sample.txt

Folder created: files

File moved to files/updated_sample.txt
Delete Error: File not found

Automation Task Completed
```